

# OIL INDIA LIMITED — DRILLING DEPARTMENT

DAILY DRILLING REPORT (DDR) · SYNTHETIC DEMONSTRATION DOCUMENT · eRTMAC-NWIS sample

|                      |            |              |            |
|----------------------|------------|--------------|------------|
| Well                 | DLJ-12     | Field        | Duliajan   |
| Rig                  | OIL-E760   | Report No.   | 114        |
| Report Date          | 2019-03-14 | Spud Date    | 2018-11-20 |
| Depth @ 24:00 (m MD) | 2,958      | Progress (m) | 64         |
| Latitude             | 27.5130    | Longitude    | 95.1486    |
| Hole Size            | 12-1/4"    | Well Type    | deviated   |

## 1. OPERATIONS SUMMARY (00:00 – 24:00 hrs)

| From  | To    | Depth (m) | Operation                                                         |
|-------|-------|-----------|-------------------------------------------------------------------|
| 00:00 | 03:30 | 2,894     | Drilled ahead 12-1/4" hole, parameters steady, monitored returns. |
| 03:30 | 04:15 | 2,916     | Connection, survey, circulated bottoms-up.                        |
| 04:15 | 09:45 | 2,912     | Torque Spike — see remarks. NPT 3.0 hrs.                          |
| 09:45 | 13:00 | 2,912     | Conditioned hole, reamed interval, circulated hi-vis sweep.       |
| 13:00 | 18:30 | 2,943     | Drilled ahead at controlled parameters.                           |
| 18:30 | 19:15 | 2,943     | Connection, flow-check, mud properties checked.                   |
| 19:15 | 24:00 | 2,958     | Drilled ahead; depth at 24:00 hrs as reported.                    |

## 2. MUD PROPERTIES

| MW (ppg) | FV (s) | PV (cP) | YP (lb/100ft²) | pH  | Solids (%) | Type        |
|----------|--------|---------|----------------|-----|------------|-------------|
| 10.5     | 73     | 28      | 30             | 9.5 | 11.2       | KCl-Polymer |

## 3. CASING STATUS

| String       | Size    | Shoe (m MD) | TOC (m) |
|--------------|---------|-------------|---------|
| Conductor    | 20"     | 76          | 0       |
| Surface      | 13-3/8" | 1,082       | 0       |
| Intermediate | 9-5/8"  | 2,952       | 2,121   |

## 4. DRILLING EVENTS / REMARKS

1. TORQUE SPIKE at 2,912 m MD (BARAIL) | Severity: MEDIUM | NPT: 3.0 hrs | MW: 10.5 ppg  
Torque increased from 9 to 13 kft-lb at 2,912 m MD on entering Barail coal seams; stick-slip observed and ROP dropped from 9 to 4 m/hr.  
Mitigation: Reduced WOB/RPM, circulated bottoms-up and added lubricant to the mud system.  
Lesson: Upper Barail coal seams produce torque spikes 20-40 m below the top; pre-condition the mud before entering.

2. STUCK PIPE at 2,940 m MD (BARAIL) | Severity: HIGH | NPT: 18.5 hrs | MW: 10.6 ppg  
Pipe became stuck while making connection at 2,940 m MD in Barail shale/coal sequence. Overpull to 85 klbs, no rotation. Worked pipe with jarring for 4.5 hrs; freed after spotting 30 bbl lubricant pill. Torque had risen from 9 to 14 kft-lb over the preceding 60 m.  
Mitigation: Increased circulation before connections; short reaming intervals; improved hole conditioning with hi-vis sweeps.  
Lesson: Rising torque trend in upper Barail is a precursor to stuck pipe; ream every stand and keep jars armed below 2,900 m.

## 5. LESSONS LEARNED

- Upper Barail coal seams produce torque spikes 20-40 m below the top; pre-condition the mud before entering.
- Rising torque trend in upper Barail is a precursor to stuck pipe; ream every stand and keep jars armed below 2,900 m.

Prepared by: Drilling Engineer (synthetic) · Reviewed by: Drilling Superintendent (synthetic) · This document was generated for the eRTMAC-NWIS demonstration and does not describe a real operation.